

Frequency Chirping in External Modulators

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Abstract—A general formula is given which expresses frequency chirping in some types of external intensity modulators, such as the loss modulator, directional coupler type modulator, Mach-Zehnder interferometer type modulator, and total internal reflection type modulator. The chirping phenomenon treated here is caused by the phase modulation due to an accompanied refractive index change. It is uniquely expressed in terms of an α -parameter which contributes to frequency chirping in the same manner as in the direct modulation of a semiconductor laser. In addition, the transmission bandwidth of a single-mode fiber system employing an external modulator is discussed by using this formula and compared with that utilizing the direct laser modulation.

I. INTRODUCTION

THE FREQUENCY chirping due to the direct modulation of a semiconductor laser limits the transmission bandwidth of a single-mode fiber system, even if we employ a directly modulated dynamic-single-mode (DSM) laser [1]. An external modulation technique is considered to be an alternative candidate in order to eliminate this problem. Some types of external intensity modulators such as directional-coupler type modulator [2]–[5], loss modulator [6]–[10], Mach-Zehnder interferometer type modulator [11], [12], and total internal reflection type modulator [13], [14], have been studied. Recently, some transmission experiments employing an external modulation technique have been reported for the purpose of eliminating the frequency chirping [15], [16]. Also, high-speed external modulators using a quantum well structure [9] and the monolithic integration of a modulator and a laser [17], [18] have been intensively studied. However, we should still consider chirp-like spectral line broadening caused by an external intensity modulation. The authors pointed out that some frequency chirping is caused by phase modulation due to a refractive index change in a loss modulator [19]. Until this moment, the dynamic spectral width in external intensity modulators has been experimentally studied [20], [21]. It seems to be very important to make clear the nature of the frequency chirping in external intensity modulators because it determines the ultimate transmission bandwidth of single-mode fiber systems employing an external modulation scheme.

In this paper, we give a general formula of the frequency chirping in external modulators. In Section II, the frequency chirping in some types of external modulators such as the loss modulator, the directional coupler type,

the Mach-Zehnder interferometer type, and the total internal reflection type, is formulated and is compared with that of a direct modulation. It is uniquely expressed in terms of an α -parameter due to the phase modulation accompanied by refractive index change. It is shown that this is the same relation expressing the frequency chirping under direct modulation. Some mathematical errors in a preliminary calculation [22] have been corrected. In Section III, the transmission bandwidth of a single-mode fiber employing an external modulator is considered by using this formula. The α -parameter newly introduced for external modulators is one of important parameters to express the performance of an external intensity modulator.

II. FORMULA OF FREQUENCY CHIRPING IN EXTERNAL MODULATORS

A. Frequency Chirping in External Modulators

In the direct modulation of a semiconductor laser, the frequency chirping is caused by the refractive index change of the active layer due to the carrier density modulation. This is expressed by the linewidth enhancement factor α [23] and the modulation waveform [24], [25]. To eliminate the chirping, an external modulation technique seems to be promising; however, chirp-like spectral broadening also must be investigated. One possible cause is the wavelength shift of the semiconductor laser due to an external reflection outside of the external modulator. This is because the coupled power of the externally reflected light to the laser cavity is varied with intensity modulation, which changes the laser wavelength. A theoretical analysis of the frequency chirping due to an external reflection has been presented [26]. This sort of chirping can be essentially removed by adopting an optical isolator and reducing the reflection by means of AR coating to the endface of the modulator. Another possible cause is the phase modulation due to the refractive index change of medium inside an external modulator, which is treated in this paper. The relation between the frequency chirping due to phase modulation and intensity modulation is derived for four types of external modulators and is shown schematically in Fig. 1.

B. Loss Modulator

One of the external modulators using semiconductor materials is a loss modulator with a variable loss (or gain) to the laser light passing through the modulator. This type of external modulator can be made with the same process as semiconductor lasers, which enables the monolithic integration of lasers and modulators. A conceptual dia-

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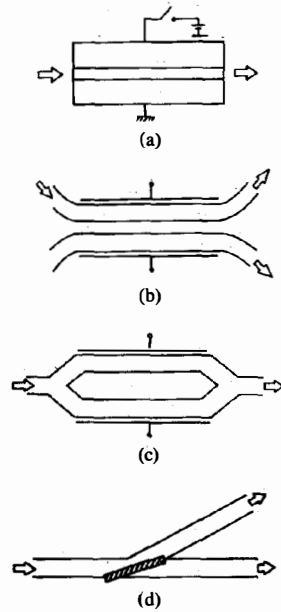


Fig. 1. Schematic structures of four types of external intensity modulators: (a) loss modulator (bulk, MQW), (b) directional-coupler type modulator, (c) Mach-Zehnder interferometer type modulator, (d) total internal reflection type modulator.

gram of an external loss modulator is shown schematically in Fig. 2. Intensity modulation can be obtained by changing the loss of a modulator. In order to change the loss in a modulator, the electroabsorption effect in quantum-well [8], [9] as well as waveguide or thin film structure bulk crystal [6], [10], or the change of injected carrier density [7] have been utilized.

In this section, let us obtain a formula for the chirping of modulated light frequency in a loss modulator. In general, if we vary the loss that is due to the imaginary part n'' of the refractive index, the real part n' will suffer some of modulation, according to the Kramers-Kronig relation. Therefore, this causes phase variation of transmitted light through an external modulator together with intensity modulation [19]. Here we will have to include the waveguide effect into the refractive index if we use a waveguide structure. In this sense, the term of equivalent index must be employed. Assuming that the input light into the modulator has a constant amplitude $|E_1|$ and constant angular frequency ω_0 , the amplitude $|E_2|$ and the relative phase change ϕ of the output are given by

$$|E_2| = |E_1| \cdot \exp(-k_0 n'' L) \quad (1)$$

$$\phi = -k_0 n' L \quad (2)$$

where k_0 is the propagation constant in free space and L is the length of an external modulator. Combining (1) and (2), we obtain the following relation which relates the instantaneous intensity ($S = |E_2|^2$) and the phase ϕ of the output light:

$$d\phi/dt = (\alpha/2)(1/S)(dS/dt) \quad (3)$$

where $\alpha = \Delta n' / \Delta n''$ is the relative change of the real

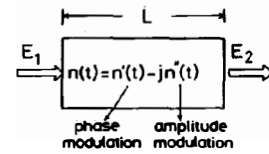


Fig. 2. Conceptual diagram of external loss modulator.

part $\Delta n'$ and imaginary part $\Delta n''$ of the refractive index. The derivative of the phase in the left side of (3) corresponds to the instantaneous angular frequency shift. It is noteworthy that this relation is identical to the linear part of that in the direct modulation of a semiconductor laser [23]–[25]. When a loss is changed by a carrier injection, α is equal to the linewidth enhancement factor derived by Henry for semiconductor lasers [23]. This value ranges from 2 to 7 for GaAs and GaInAsP conventional lasers [27]–[32] and from 1.5 to 1.7 for a quantum well laser [33]. Therefore, the external modulator using the same process as lasers gives almost the same frequency chirping as the direct modulation, whereas ringing of carriers due to relaxation oscillation in external modulation is not noticeable as in lasers and the chirping can be reduced by choosing some suitable waveforms of modulation signals. Suzuki *et al.* obtained the α -parameter for the electroabsorption type bulk modulator, which was 1–2 for a discrete modulator [20] and 1.6 for an integrated modulator with a DFB laser [18]. Also, the α -parameter for a loss modulator using a quantum-well structure was experimentally estimated by Wood *et al.* to be 0.9 [21]. Theoretically, Nagai *et al.* show that the α -parameter in a quantum well structure depends on the operating wavelength and applied electric field [34].

C. Directional Coupler Type

Optical switching devices or modulators built by electrically switching a directional coupler from the crossover state to the straight-through state have been widely studied [2]–[5]. However, the chirp-like spectral broadening due to phase modulation has not yet been discussed. Their switching characteristics can be analyzed by using the coupled mode theory [35]. A preliminary calculation of the frequency chirping in this type of modulator was presented by the present authors [22]. We found some mathematical error in that earlier calculation and have corrected it and pointed out the existence of bias dependency here. The theoretical model is shown schematically in Fig. 3. Only the waveguide 1 is assumed to be initially excited by the light with a constant amplitude and constant frequency. The complex amplitudes of the electric fields E_1 and E_2 in two waveguides 1 and 2 at the output ends are derived as follows [35]:

$$E_1 = E_0 \cdot \left[(\kappa^2 \cos^2 \{ (\kappa^2 + \Delta\beta^2)^{1/2} L \} + \Delta\beta^2) / (\kappa^2 + \Delta\beta^2) \right]^{1/2} \cdot \exp(-j\bar{\beta}L + j\phi_1) \quad (4)$$

$$E_2 = E_0 \cdot j\kappa \sin \{ (\kappa^2 + \Delta\beta^2)^{1/2} L \} / (\kappa^2 + \Delta\beta^2)^{1/2} \cdot \exp(-j\bar{\beta}L + j\phi_2) \quad (5)$$

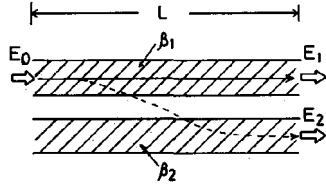


Fig. 3. Schematic structure of directional-coupler type modulator.

where E_0 is the amplitude of the electric field at the input, κ is a coupling coefficient, L is a device length, and

$$\Delta\beta = (\beta_1 - \beta_2)/2 \quad (6)$$

$$\bar{\beta} = (\beta_1 + \beta_2)/2 \quad (7)$$

with the propagation constant of β_i in the waveguide i . When $\bar{\beta}$ is assumed to be constant, the phase of the output light ϕ_1 and ϕ_2 are given by

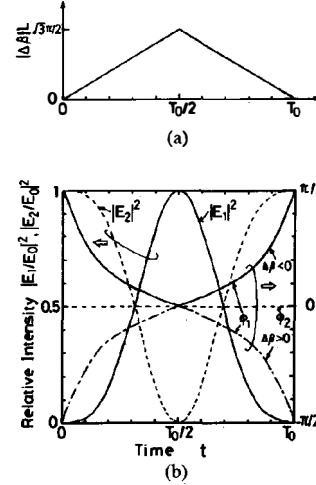
$$\phi_1 = -\tan^{-1} \left[\frac{\Delta\beta \sin \{(\kappa^2 + \Delta\beta^2)^{1/2} L\}}{(\kappa^2 + \Delta\beta^2)^{1/2} \cos \{(\kappa^2 + \Delta\beta^2)^{1/2} L\}} \right] \quad (8)$$

$$\phi_2 = 0. \quad (9)$$

The output intensity of both waveguides is modulated by varying $\Delta\beta$. $\Delta\beta$ is varied by changing the waveguide indices due to the electrooptic effect. When the device length L is equal to the coupling length given by $\pi/2\kappa$, all the energy is transferred from waveguide 1 to waveguide 2. Now, we considered the phase modulation accompanied by the intensity modulation. As can be seen in (8) and (9), the phase of the output light in waveguide 2 is constant when $\bar{\beta}$ is fixed. Therefore, the intensity of the light which has been coupled from the waveguide initially excited can be modulated without phase modulation, when the propagation constant of the two waveguides changes by the same amount $\Delta\beta$ but in the opposite sign. This is due to the compensation of the phase shift of the coupled light. On the other hand, the phase of the light straight-through the waveguide initially excited changes together with the intensity modulation. We considered a symmetric sawtooth modulation of $\Delta\beta$. Fig. 4 shows the calculated relative output intensity $S_i = |E_i/E_0|^2$ and the corresponding phase ϕ_i of both waveguides ($i = 1, 2$) against time by using (4)–(9). The sign of the derivative of the phase with respect to time in waveguide 1, which corresponds to the instantaneous angular frequency shift, depends on the sign of $\Delta\beta$. Therefore, both cases of “red-shift” and “blue-shift” chirping possibly occur. To make clear the nature of the frequency chirping in a directional coupler type modulator, we simply formulate the chirping. We can find that (3) means that the α -parameter is the ratio of the amplitude modulation and the phase modulation, namely

$$\alpha = (d\phi/dt) / ((1/E)(dE/dt)) \quad (10)$$

where E and ϕ are the amplitude and phase of an electric field, respectively. We may calculate the α -parameter de-


 Fig. 4. (a) modulation shape of $\Delta\beta$, (b) changes of intensity and corresponding phase at the output ends of two waveguides.

finied by (10) for the directional coupler type modulator by using (4), (5), (8), and (9), as follows:

$$\alpha = -\frac{\Delta\beta^2 \gamma^2 L + \kappa^2 \gamma \sin(\gamma L) \cos(\gamma L)}{\kappa^2 \Delta\beta \sin(\gamma L) \{ \sin(\gamma L) - \gamma L \cos(\gamma L) \}} \quad (11)$$

(for waveguide 1)

$$\alpha = 0 \quad (12)$$

(for waveguide 2)

where $\gamma^2 = \kappa^2 + \Delta\beta^2$. Thus, the chirping in a directional coupler type modulator can be also expressed by

$$d\phi/dt = (\alpha/2)(1/S)(dS/dt)$$

which is the same equation as a loss modulator in terms of the α -parameter defined by (10). Equation (11) shows the α -parameter is a function of $\Delta\beta$, which indicates α depends on the bias level of a relative output intensity $S_b = (|E_1/E_0|^2)$ of waveguide 1. Fig. 5 shows the calculated α -parameter for waveguide 1 as a function of the bias intensity S_b . The sign of α is determined by the sign of $\Delta\beta$. Although the α -parameter depends on S_b , α is nearly equal to 1 when the bias level of S_b is set to a half of the input intensity. On the other hand, α is zero for waveguide 2. This result can be applied to all directional coupler type modulator, regardless of size and material.

D. Mach-Zehnder Interferometer Type

Frequency chirping of a Mach-Zehnder interferometer type modulator with the device length of L shown in Fig. 6 is also considered. The output electric field E is given by

$$\begin{aligned} E &= E_0/2 \exp(-j\beta_1 L) + E_0/2 \exp(-j\beta_2 L) \\ &= E_0 \cos(\Delta\beta L) \exp(j\bar{\beta} L) \end{aligned} \quad (13)$$

where E_0 is the amplitude of the electric field in the input waveguide, $\Delta\beta$ and $\bar{\beta}$ are given by (6) and (7), respectively. We find that the phase modulation can be completely compensated if the propagation constants of two

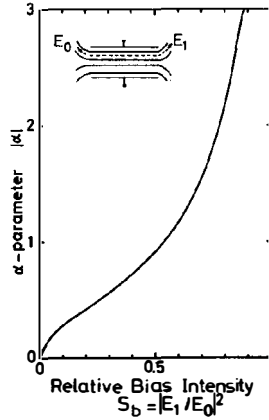


Fig. 5. Calculated α -parameter for straight-through light as a function of the relative bias intensity S_b at the output of the waveguide 1.

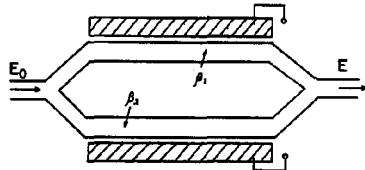


Fig. 6. Schematic structure of Mach-Zehnder interferometer type modulator.

waveguides are changing by the same amount $\Delta\beta$ with an opposite sign. Thus, the α -parameter defined in (10) in this type of modulator is equal to zero.

E. Total Internal Reflection Type

The principle of a total internal reflection type modulator shown in Fig. 7 is that a reflected light or a transmitted light is modulated by varying the amplitude reflectivity at the boundary AB due to refractive index change in region II in Fig. 7. We consider the phase modulation in the reflected light and transmitted light, assuming that they are plane waves for simplicity. The amplitude reflectivity is given by

$$r = (n_1 \cos \theta_2 - n_2 \cos \theta_1) / (n_1 \cos \theta_2 + n_2 \cos \theta_1) \quad (14)$$

where θ_1 and θ_2 are the incident and refracted angle, respectively, and n_1 and n_2 are the effective indices in region I and II, respectively. From (14), we can see that the phase shift at the reflection is constant and equal to π when the incident angle is larger than or equal to the critical angle. Therefore, the intensity of the reflected light can be modulated without any phase modulation. In other words, the α -parameter is equal to zero. On the other hand, the transmitted light suffers some of refractive index change. By calculating the optical pass length, the phase shift of the transmitted light is

$$\begin{aligned} \Delta\phi &= k_0 n_2 d / \cos \theta_2 - k_0 n_1 d \sin(\theta_1 - \theta_2) / \cos \theta_2 \\ &= k_0 d (n_2 \cos \theta_2 - n_1 \cos \theta_1). \end{aligned} \quad (15)$$

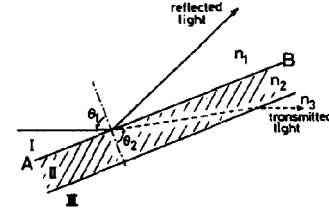


Fig. 7. Conceptual diagram of total internal reflection type modulator.

Thus we can find $|\Delta\phi| < k_0 n_1 d \cos \theta_1$. Usually, a possible refractive index change is so small then $\cos \theta_1 \ll 1$, and also the width d of region II is assumed to be in the order of wavelength (note that d must be larger than the penetration depth for total internal reflection). Consequently, the frequency shift is much less than the modulation angular frequency ω_m , namely

$$|d\phi/dt| \cong |\Delta\phi/T| \ll 2\pi/T = \omega_m \quad (16)$$

where T is a modulation period. This means that the spectral broadening due to the phase modulation is negligible, namely $|\alpha| \ll 1$.

Rough estimation of frequency chirping in total internal reflection type modulator was carried out here. However, we should note that the phase shift of the reflected light changes when the incident angle is less than the critical angle. Therefore, when we consider this type of modulator with three dimensional waveguide where the incident light consists of multiple plane waves with different propagating directions, some phase modulation is caused. Detailed analysis is still open to investigation.

F. Discussion

The frequency chirping in four types of the external intensity modulators treated here can be summarized by the same equation which simply uses the α -parameter, that is, the quantity of the chirping is characterized by

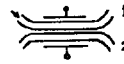
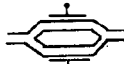
$$d\phi/dt = (\alpha/2)(1/S)(dS/dt)$$

in terms of the α -parameter. Table I shows the α -parameter for four types of modulators in comparison with the direct modulation of a semiconductor laser. The α -parameter for a loss modulator employing the carrier injection may be almost same as direct laser modulation, which depends on operating wavelength and is ranging from 2 to 7 [36]. On the other hand, that of a loss modulator using the electroabsorption in a bulk and quantum well experimentally exhibited smaller value ($|\alpha| \cong 1$) [20], [21]. Theoretically, it depends on operating wavelength and the applied electric field [34] and therefore the optimization of device structure and operating condition may reduce the frequency chirping. In principle, the frequency chirping of directional coupler type, Mach-Zehnder interferometer type and total internal reflection type modulators can be eliminated.

III. RELATED TRANSMISSION BANDWIDTH

The frequency chirping in an external modulation affects the transmission properties of a single-mode fiber as

TABLE I
 α -PARAMETER FOR VARIOUS TYPES OF EXTERNAL MODULATORS IN
 COMPARISON WITH DIRECT MODULATION

Semiconductor Laser		Conventional: 26)-30) $\alpha=2-7$ QW: $\alpha=1.5-1.7$ 31)
Loss Modulator	Electro-Absorption 	Bulk: $ \alpha =1-2$ 18), 20) depending on λ, E QW: $ \alpha =1$ 21) depending on λ, E
	Carrier Injection 	$\alpha=2-7$ 35) depending on λ
Electro-Optic Modulator	Directional Coupler 	$ \alpha =1$ for waveguide:1 depending on intensity $\alpha=0$ for waveguide:2
	Mach-Zehnder Interferometer 	$\alpha=0$
	Total Internal Reflection 	$\alpha=0$

well as in a direct modulation. In Section II, we found that the frequency chirping in an external modulation is expressed by the same equation as in a direct modulation. By using this result, we can obtain the transmission bandwidth of a single-mode fiber of length L employing an external modulator in the same manner of [25]. The following dispersion formula is used to evaluate the bandwidth [37]:

$$E_{\text{out}}(t) = \int_{-\infty}^{\infty} \epsilon(\omega) \exp [j\{\omega t - \beta L - \beta'(\omega - \omega_0) - (\beta''/2)(\omega - \omega_0)^2\}] d\omega \quad (17)$$

where E_{out} is the electric field at the fiber end, β , β' , and β'' are the propagation constant, its first and second derivatives of the fiber with respect to the angular frequency, and $\epsilon(\omega)$ is the spectrum of externally modulated light. Now we consider the pulse response of a single-mode fiber, assuming the input pulse has a Gaussian shape. In the same procedure as in [25], we obtain the maximum corresponding bit rate B as follows, which is defined by the reciprocal of the attainable minimum output pulsewidth:

$$B = 1/[2\{2(\alpha^2 + 1)^{1/2} + 2\alpha\}^{1/2} |\beta''L|^{1/2}]. \quad (18)$$

Fig. 8 shows the calculated product of the maximum bit rate B and the square root of the fiber length L as a function of α parameter, where a wavelength and a fiber dispersion are assumed to be 1.55 μm and 20 ps/km/nm, respectively. In the range of the α -parameter from -4 to 0 , the transmission bandwidth exceeds the value which is limited by the sideband of the modulation signal, which is due to the pulse compression caused by "blue shift chirp."

The modulation waveform is also important for the

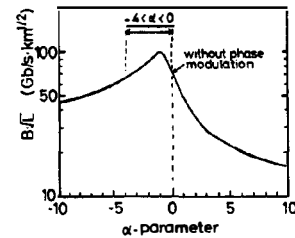


Fig. 8. Product of maximum bite rate and the square root of the fiber length as a function of the α -parameter in an external modulator.

transmission characteristic. The change of the intensity causes the chirping according to (3). Therefore, we can see that the reduction of the transient time in a pulse eliminates the chirping. As described in [19] and [38], the chirping free transmission can be achievable by using rectangular shape modulation form. Therefore, wide-band external modulation with optimized pulse shape can be effective to reduce the effect of the chirping.

IV. CONCLUSION

Frequency chirping effect in some types of external intensity modulators, such as a loss modulator, directional coupler type modulator, Mach-Zehnder interferometer type modulator and total internal reflection type modulator can be uniquely expressed by an equation in terms of α -parameter as well as direct laser modulation. The α -parameter which determines the amount and sign of the frequency chirping, is one of the important parameters expressing the performance of high-speed external modulators. By using this simple formula, we can easily analyze the frequency chirping accompanied by intensity modulation and its effect on the single-mode fiber transmission. The determination of α -parameter for various devices is still open to investigation.

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